

Programmable Sensor for On-Line Checking of Signal Integrity in FPGA-Based Systems Subject to Aging Effects

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Abstract—In current nanometer technologies, aging effects (due for instance to Negative Bias Thermal Instability) may appear after relatively short operating times, compared to the expected lifetime of circuits, even for relatively short-cycle consumer electronics. Therefore, there is an increasing need for on-chip aging monitoring. This paper presents a programmable aging sensor that can be embedded in FPGA-based designs, using standard resources available in those devices. The sensing principle is to monitor performance degradation over time. Depending on whether dynamic or static aging effects are dominant, the sensor can operate continuously or be only activated at some time intervals. Given the reduced amount of resources required by the sensor, it can be instantiated not only in the critical paths of a circuit, but also in those that may be identified to be more likely affected by aging effects. Experimental results are presented to demonstrate the performance of the proposed sensor.

Keywords—aging sensor, delay faults, Field Programmable Gate Array, observation interval

I. INTRODUCTION

The constantly increasing uncertainty of circuits' operation as nanometer technologies scale down brings new challenges to design and test strategies, as well as new defects and unsolved problems. The high variability in small gate length technologies is caused by the increased dependence of path delay on many variables [1] and this is becoming one of the leading causes for chip failures and delayed schedules [2].

In general, circuits' signal integrity is affected by phenomena such as MOS threshold voltage (V_{th}) variations, thickness variations [3], gate length and width variations, crosstalk [4], IR-drop [5], Process, power supply Voltage, and Temperature (PVT) variations [6], and long-term effects, such as Negative Bias Thermal Instability (NBTI) [6] or radiation, any of which may cause delay errors. With technology scaling down, all these effects are amplified, uncovering circuits' instabilities. Therefore, new challenges arise to improve design and test of new node's digital circuits and regain circuits' *dependability* (defined as the trustworthiness of the circuit, or the confidence on the correct circuit behavior within the specified time frame).

Considering semiconductor aging [8], it causes long-term performance degradation with effects that are cumulative in time, which degrade continuously circuit's dependability and reliability, with variability always increasing. From literature, many long-term effects can be identified to cause this long-term performance degradation. For instance, V_{th} parametric variations may be induced either due to holes trapped in the thin gate oxide (SiO_2), or due to Si/SiO_2 surface change. The first effect tends to decrease V_{thn} and to increase $|V_{thp}|$ (NMOS and PMOS threshold voltages, respectively). The second effect, under high radiation, tends to increase V_{thn} and $|V_{thp}|$ [9]. Hence, as time elapses, V_{th} shifts will modify the timing performance of digital circuits. Long term radiation effects are usually characterized by TID (Total Ionizing Dose), which primarily affects NMOS device characteristics [10][11]. TID also increases leakage under isolation oxides and at MOS gate edges. TID mitigation is object of intense research [12].

The most important long-term effect causing the reduction of circuits' deterministic behavior over time is the temporal reliability degradation in MOSFETs, the NBTI effect [6][13]. It affects primarily the PMOS transistors, increasing $|V_{thp}|$ over time. This will ultimately cause the de-synchronization of memory cells, identified as delay faults. The impact of NBTI on digital circuit performance has been under research [6][13][14] and NBTI modeling in static and dynamic operation has been also proposed [14]. Solutions to make circuits more tolerant to V_{thp} degradation and to restrict NBTI performance degradation have been presented in [13][15][16]. More general solutions to increase circuit's dependability for aging effects were also presented in [17]–[19], with various aging sensor topologies aimed at globally detect circuit's performance degradation. In fact, aging sensors are being subject to important research works, and improved aging sensors' topologies can predict circuits' functionality failure, by locally (where the delay faults actually occur, in the memory cells) monitoring performance degradation [20]–[25].

The increased usability of FPGAs to replace ASICs in low-volume applications, and the use of new node technologies to produce today's FPGA devices, impose the need to adapt traditional hardware solutions to improve circuit's dependability, including the implementation of aging sensor

functionality on these devices. Therefore, the purpose of this work is to present an implementation of the functionality of a hardware aging sensor approach [23]-[25] in FPGAs, defining a new aging sensor topology for reconfigurable systems.

The paper is organized as follows. In Section II the proposed aging sensor is presented. In section III, experimental results are presented for two test circuits. Finally, section IV summarizes the main conclusions and provides guidelines for future work.

II. PROPOSED SENSOR

The principle of operation of the proposed sensor is based on the definition of an observation interval (T_g) in each clock period, close to the active clock edge, whose duration is programmable, with a resolution (that depends on the operating frequency) in the range of tens of picoseconds. The sensor checks for the occurrence of transitions of the target signals (those whose integrity is more likely to be compromised by aging affects) within the observation interval. In normal operation, the signal is fast enough, so transitions do not take place during T_g . As the signal becomes slower because of aging effects, the probability of a transition occurring within T_g increases. Whenever such a transition takes place, the sensor flags it, indicating that transitions are occurring too close to the active clock edge, and may thus cause signal integrity loss. The above mentioned resolution of the sensor is suitable for these purposes, because in most designs transitions should not be allowed to occur only a few picoseconds before the active clock edge.

In this work, the proposed sensor (Figs. 1 and 2) has been implemented in a 90-nm Spartan-3AN FPGA from Xilinx. As explained below, it uses only (a few) standard FPGA resources and, therefore, it can also be implemented in other devices from the same vendor (and, with minor modifications, in devices from other vendors).

In Fig. 1, the critical path of the target circuit terminates at the input of flip-flop FF2. Two Digital Clock Managers (DCMs) are in charge of generating the circuit's main clock signal (clock1) and an auxiliary clock signal (clock2) with the

same frequency, but exhibiting a *negative phase shift* equal to T_g , with regard to clock1. Therefore, clock1 is in fact delayed T_g with regard to clock2. For a given clock period T_{clk} , DCMs allow T_g to be configured to take any value $T_g = N * T_{clk} / 256$, where N is an integer in the range $-255 < N < 255$. DCMs are standard resources available in current FPGAs [26], which, among other functionalities, allow such negative phase shifts to be applied to clock signals. The reason why this solution has been adopted, instead of directly delaying clock2 to obtain clock1, is to keep the main clock unchanged with regard to the way the vendor synthesis tools generate it.

It is clear from Figs. 1 and 2 that FF3 captures the data in the critical path a time T_g before FF2 does. Therefore, if the outputs of FF2 and FF3 differ, it follows that a transition of the signal took place during T_g . In order to check for this situation, the Decision Circuit (DC) captures the outputs of FF2 and FF3 in another two flip-flops (FF4 and FF5, respectively) activated by the falling edge of clk2. This means that $T_g < T_{clk}/2$, but this is clearly not a real limitation in practice. The DC circuit has two outputs, one combinational and the other registered. The latter continuously flags the occurrence of a transition of the target signal during T_g , until FF6 is reset by a dedicated signal. The Look-Up Table (LUT) in Fig. 2 implements an XOR function, whose output equals "1" at least during one clock cycle upon the occurrence of a transition during T_g .

As long as a given circuit is powered up, aging effects can affect both areas with low activity (or inactive) and with high activity. Depending on the technology, static or dynamic aging effects can be more significant. Aging sensors should be as robust as possible to aging effects, which would impact their timing performance. Taking this into account, the proposed sensor includes an on/off signal, which allows designers to decide whether it will be operating continuously (in case static aging is more significant) or disconnected, except during short time intervals where aging effects are evaluated (in case dynamic aging is more significant).

In order for the proposed sensor to work properly, the delays from the starting node of the critical path to the inputs of FF2 and FF3 must be the same, which means that $t_{d1} = t_{d2}$. This

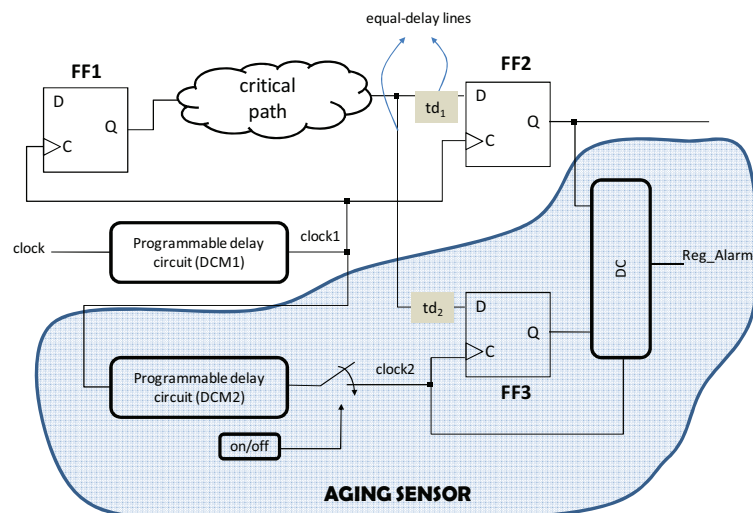


Figure 1. Block diagram of the proposed aging sensor

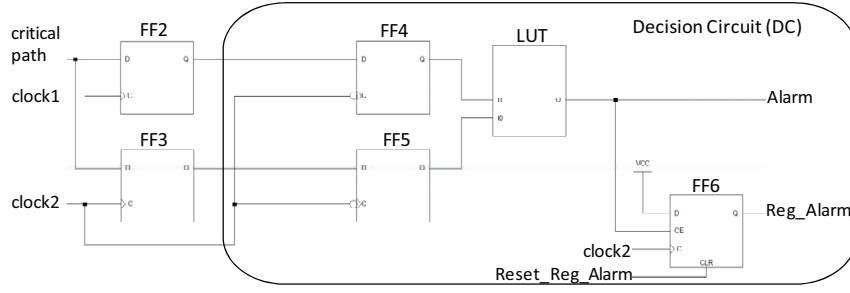


Figure 2. Decision circuit

condition can be accomplished in FPGAs, by placing both FF2 and FF3 in slices of the same Configurable Logic Block (CLB, the basic logic building block of FPGAs [26]). This imposes a limitation, easy to comply with in most cases, to the placement of FF3. Interestingly, all remaining elements of the sensor can be freely placed in any available area of the device.

The sensor requires only very few (and standard) FPGA resources: one DCM, one LUT, and 3 flip-flops. Therefore, it is possible to include several sensors if required by a given circuit, because of the existence of more than one critical path or of areas that are identified to be more prone to be affected by aging effects. The main limitation is the number of DCMs available in the FPGA used. However, these could be shared among different sensors. In addition, advantage can be taken of the dynamic reconfiguration capabilities of some FPGAs to place sensors in different areas of the FPGAs during different time intervals. Given its very low complexity, the impact of the sensor in power consumption can be assumed to be not significant in most cases.

As the sensor is located at the output of critical paths, the effect of this increased load in the critical delay needs to be carefully evaluated. This effect is circuit-dependent, mainly because of interconnects. In the two circuits where the experimental results presented in next section have been obtained, the inclusion of the sensor caused a small increase of the critical delay of 30 and 39ps, respectively.

One very important characteristic of the proposed sensor is that its behavior is insensitive to power supply variations in the FPGA core (to the limit that causes functional collapse of the device). The critical issue in this regard is the matching of the delays t_{d1} and t_{d2} which, as mentioned above, is guaranteed by construction. This fact is not only important from the point of view of the correct operation of the sensor, but also for validation purposes, because it is possible to emulate the effect of aging in the delays of the target circuit by decreasing the power supply voltage, without affecting the operation of the sensor.

III. EXPERIMENTAL RESULTS

Experiments using actual FPGA devices have been carried out to characterize the relationship between the increases of the delay of the critical path of a given circuit and the values of T_g that detect them.

As discussed in the previous section, the behavior of the sensor is not influenced by power supply variations affecting the core of the FPGA. Advantage has been taken of this fact

and of the similar effects of aging and power supply reduction on signal propagation delays (both cause them to increase) to emulate the occurrence of aging effects by decreasing the power supply voltage of the FPGA core. In all experiments, the circuits operate at 50MHz, resulting in a timing resolution of $20\text{ns}/256=78.125\text{ps}$ for T_g . The nominal value of the power supply voltage of the FPGA core is 1.2V. The critical paths and the corresponding delays in nominal conditions are obtained from the vendor design tools.

Propagation delay variation models previously proposed by the authors [27] have been used to relate the variations of the power supply voltage (ΔV_{DD}) with the variations of the propagation delay (ΔT_{pd}) of the critical path.

The behavior of the sensor has been first tested using the simple circuit in Fig. 3. The corresponding ΔT_{pd} (ΔV_{DD}) model is depicted in Fig. 4, and the obtained measurements are shown in Table I. De-synchronization of the circuit occurs at $V_{DD}=0.969\text{V}$ (denoted as V_{DD_fault}). In Fig. 5 it can be noticed that, in the operational range of the sensor, ΔT_{pd} (ΔV_{DD}) exhibits a linear response. Fig. 6 shows the (also linear) relationship between the V_{DD} value at which the sensor should flag the alarm signal, which represents the threshold value of timing (i.e., aging) effects, and the corresponding value of T_g that has to be configured in the sensor. It can be seen that larger values of T_g imply a higher sensitivity of the sensor, as smaller V_{DD} depletions (less aged circuit) are flagged. It can be also clearly concluded from Figs. 5 and 6 that the behavior of the sensor exhibits a very high linearity.

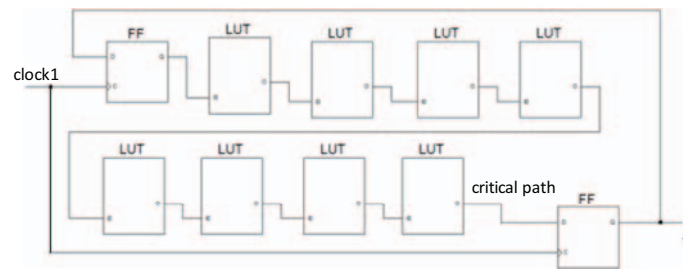


Figure 3. Test circuit #1

A more complex circuit (Fig. 7), consisting of the connection of the ISCAS c432 benchmark to a combinational cone of a fleet management system (XTRAN) from Tecmic has been used to characterize the behavior of the sensor in a more complex scenario. In this case, the impact of the inclusion of the sensor has also been analyzed. The ΔT_{pd} (ΔV_{DD}) models, with and without sensor, are shown in Fig. 8 (a) and (b) respectively. In these figures, ΔV_{DD} values at which the models

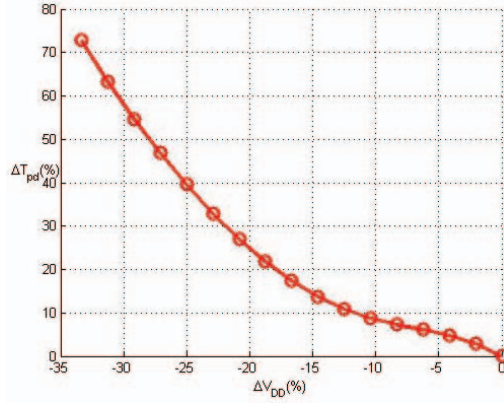


Figure 4. Delay variation model for the critical path of Fig. 3.

TABLE I. MEASUREMENTS FOR THE CIRCUIT IN FIG. 3

N, T _g (ps)	V _{DD_alarm} (V)	ΔV _{DD} (%)
-1 (78.125)	0.972	-19
-2 (156.25)	0.975	-18.75
-3 (234.375)	0.978	-18.5
-4 (312.5)	0.981	-18.25
-5 (390.625)	0.985	-17.92
-6 (468.75)	0.988	-17.67
-7 (546.875)	0.991	-17.42
-8 (625)	0.994	-17.17
-9 (703.125)	0.997	-16.92
-10 (781.25)	1	-16.67

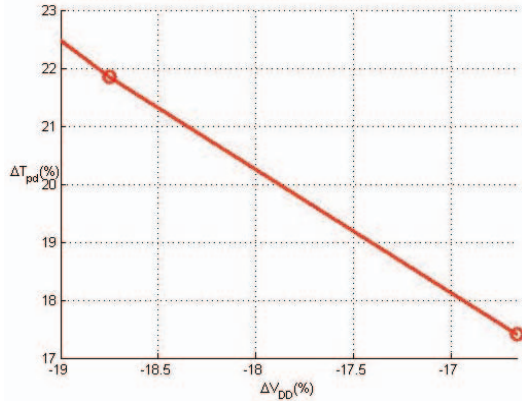


Figure 5. Operating range of the proposed sensor for the circuit in Fig. 3, according to Table I.

predict synchronism to be lost are marked with an “x”. They correspond to $V_{DD_fault}=0.805V$ and $0.772V$, respectively, whereas the actual values measured were $0.793V$ and $0.766V$, respectively, very close to predicted ones.

The more robust behavior of the circuit including the sensor with regard to ΔV_{DD} is due to the different synthesis results provided by the vendor design tool, which resulted in smaller delays when the sensor is included. This fact points to the need to carefully evaluate the locations of the sensor in a given circuit with multiple critical paths. It must be noted that, as

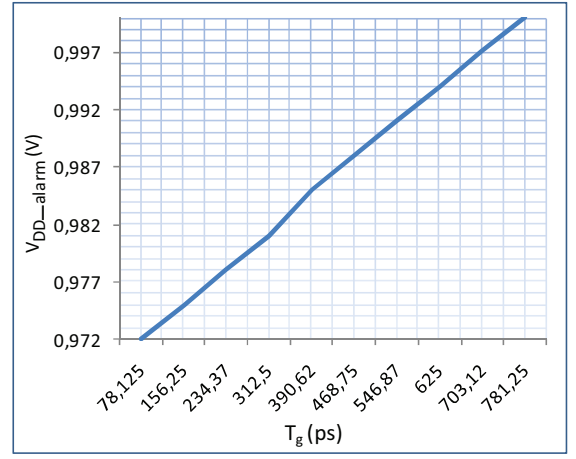


Figure 6. Relationship between V_{DD} at which the aging alarm is flagged, V_{DD_alarm} , and T_g , for the circuit in Fig. 3

several DCMs are available in current FPGAs, critical paths and others identified to be more likely affected by aging effects can be simultaneously addressed.

The measurements for test circuit #2 are shown in Table II. These results can be also seen in graphical form in Figs. 9 and 10, where the linearity of the sensor is again clearly highlighted. Note that, as the two test circuits are operating at the same frequency but their critical delays are different, their time slacks are also different, and so are the operating ranges of the sensor in each case.

Finally, the waveforms corresponding to the actual behavior of the circuit in Fig. 7 working at nominal V_{DD} , at $\Delta V_{DD}=-35.83\%$, with $T_g=78.125ps$, and at the V_{DD} at which the circuit fails ($V_{DD_fault}=0.766V$) are shown in Fig. 11. In Fig. 11(b), the combinational output of the sensor is active during one clock cycle, indicating that a transition (“0”→“1” in this case) took place in the critical path during T_g , i.e., too close to (less than 78.125ps before) the active clock edge. Note that this pulse is not present in Fig. 11(c) because loss of synchronism implies that the transition of the signal in fact takes place in the next clock cycle, i.e., outside T_g . This is the reason why the registered signal must be used as output of the sensor.

IV. CONCLUSION

In this paper, a new aging sensor for FPGA-based circuits has been presented. Its operating principle is based on an observation interval (T_g). If signal transitions occur during this interval, it follows that the delays of the circuit have increased to an unacceptable extent because of aging effects. The proposed sensor is not affected by power supply voltage variations, and it can be made more robust to either static or dynamic effects by having it working at some time intervals or continuously. The sensor is composed of a few standard FPGA resources, which make it suitable for implementation in different devices, and allows it to be instantiated several times in a circuit if required. Experimental results have been presented and discussed, demonstrating the performance of the proposed solution.

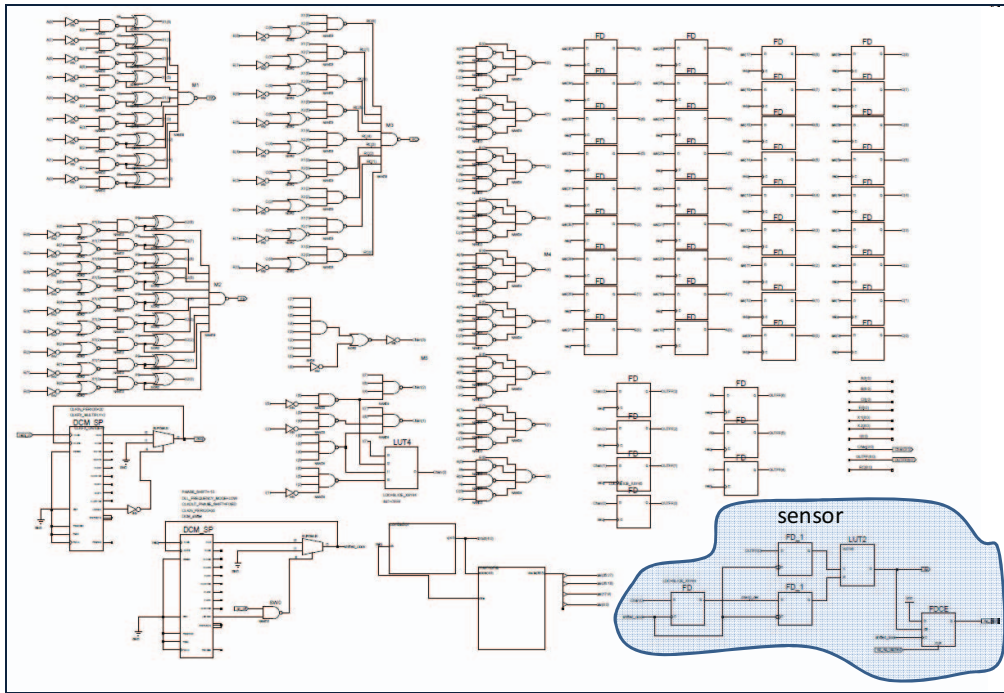
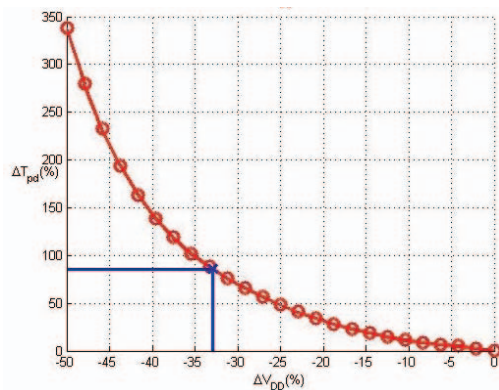
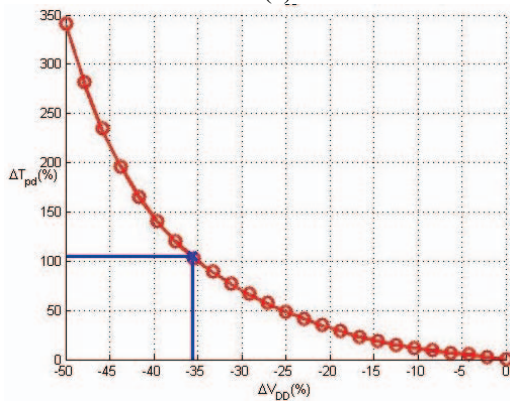


Figure 7. Test circuit #2



(a)



(b)

Figure 8. Delay variation model for the critical path in Fig. 7: (a) without aging sensor, (b) with aging sensor

As a future work, the robustness of the sensor to temperature and process variations needs to be further investigated. Also, aging experiments will be carried out to

determine the dominant effect (static or dynamic) on the target devices, the relationship between aging effects and the required values of T_g , and the sensitivity of the sensor to aging.

TABLE II. MEASUREMENTS FOR THE CIRCUIT IN FIG. 7

N, T_g (ps)	V_{DD_alarm} (V)	$\Delta V_{DD}(\%)$
-1 (78.125)	0.770	-35.83
-2 (156.25)	0.771	-35.75
-3 (234.37)	0.772	-35.66
-4 (312.50)	0.774	-35.50
-5 (390.625)	0.776	-35.33
-6 (468.75)	0.777	-35.25
-7 (546.87)	0.778	-35.16
-8 (625.00)	0.779	-35.08
-9 (703.12)	0.781	-34.91
-10 (781.25)	0.783	-34.75

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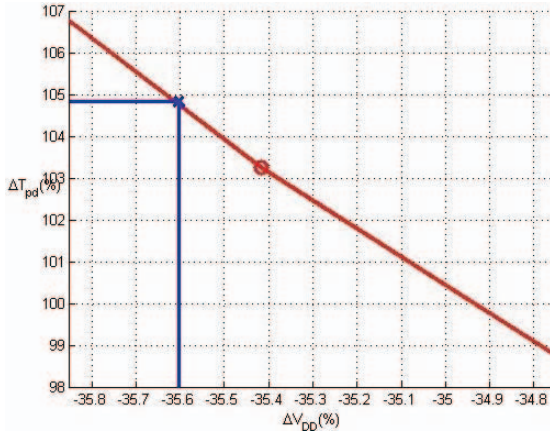


Figure 9. Operating range of the proposed sensor for the circuit in Fig. 7, according to Table II.

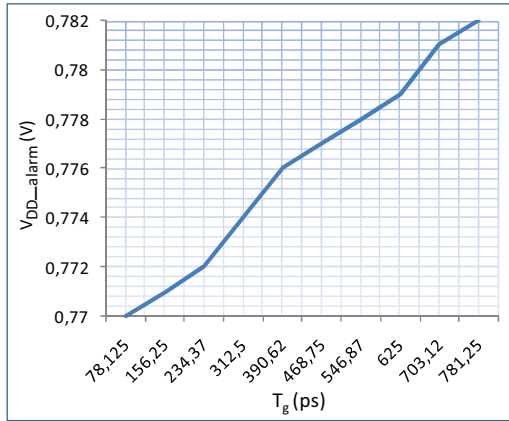


Figure 10. Relationship between V_{DD_alarm} and T_g , for the circuit in Fig. 7

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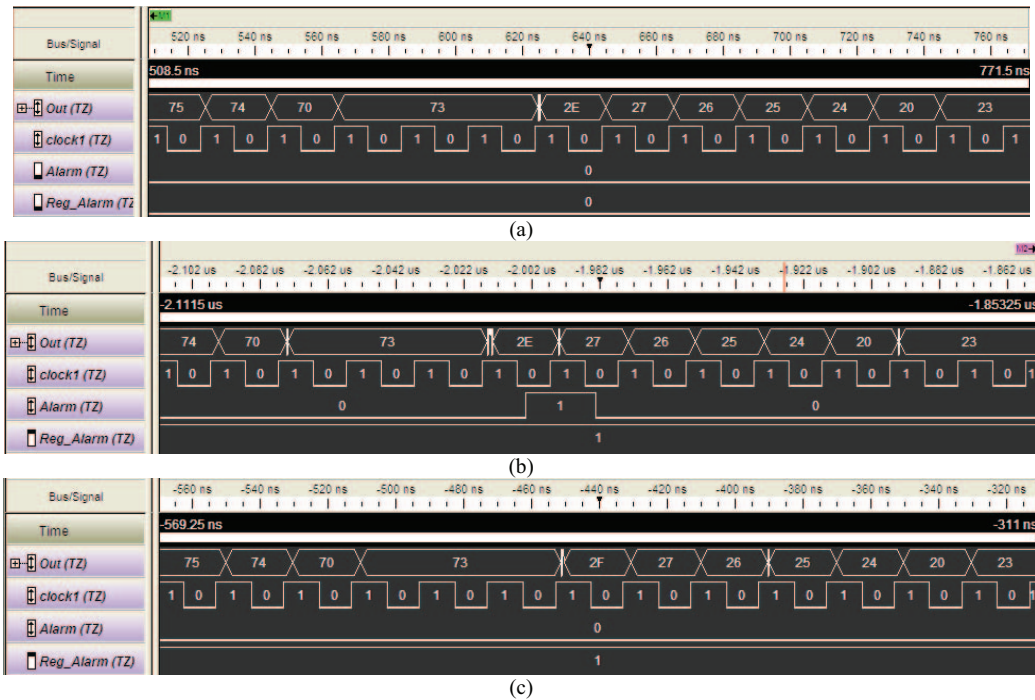


Figure 11. Waveforms of the circuit in Fig. 7 (a) at nominal (1.2V) V_{DD} (b) at V_{DD_alarm} (0,769V) for $T_g=78,125ps$, (c) at V_{DD_fault} (0,766V)

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